

JULIAN TANUI

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[Portfolio](#)

PROFESSIONAL SUMMARY

Software and AI Engineer holding a BSc in Informatics and Computer Science (Second Class Honours, Upper Division) from Strathmore University, currently building production software for the climate technology sector at Verst Carbon. My work spans full-stack web applications, AI agent systems, and data pipelines, from geospatial decision-support dashboards to a real-time meeting intelligence platform and a deepfake detection system. Recognized for academic excellence through the Dean's List, I focus on turning complex problems into reliable, well-crafted products.

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, Kotlin, SQL.
- **Frameworks:** Next.js, React, FastAPI, Flask, Node.js/Express, PyTorch, TensorFlow.
- **AI & Agents:** LLM agents (LangGraph, LangChain, multi-agent orchestration), model-selection routing across Claude/GPT-4o/Gemini, prompt engineering & evaluation frameworks.
- **Retrieval & ML:** RAG (FAISS, BM25, pgvector, ontology-backed retrieval), vector stores & semantic search, Computer Vision (CNNs, ResNet50), NLP.
- **Data Engineering:** ETL/ELT pipelines, data-quality/validation & freshness monitoring, geospatial sources (NASA FIRMS, GFW, GDELT, OpenStreetMap), interactive map dashboards.
- **Cloud & DevOps:** Docker, CI/CD, PostgreSQL/Supabase, Redis, MinIO, Modal, Inngest, Railway, Cloudflare Pages.
- **Tools:** Git/GitHub, Linux Systems, Cisco Networking, IoT Sensors.

EDUCATION

Strathmore University, Nairobi | Bachelor of Informatics and Computer Science

Graduated August 2026 | Second Class Honours, Upper Division

- **Honors:** Dean's List Award (2024-2025, 2025-2026) – Recognized for outstanding academic performance.
- **Leadership:** Local Committee Vice President for Finance and Legalities; Project Manager – SDG Hub, Strathmore.
- **Relevant Coursework:** Data Structures and Algorithms, Software Engineering, Machine Learning, Advanced Database Systems, Advanced Networking, Project Management.

PROFESSIONAL EXPERIENCE

Verst Carbon | SWE/AI Engineering Intern | Feb 2026 – Present

- Designed and built multi-agent AI systems (LangGraph, LangChain) that automate multi-step analysis across geospatial, infrastructure, economic, and monitoring data, routing each task to the most suitable language model for accuracy and cost efficiency.
- Applied retrieval-augmented generation to ground AI responses in trusted sources, keeping outputs accurate and traceable.
- Built full-stack web applications with Next.js, React, and Python (FastAPI, Node.js), including authenticated dashboards, interactive maps, and real-time chat.

- Developed data pipelines that ingest and validate large geospatial and economic datasets, with automated quality and freshness checks.
- Containerized and deployed services with Docker and CI/CD, adding monitoring and evaluation to track reliability.

CIC Insurance Group | ICT Intern | Jan 2025 – April 2025

- Provided enterprise-level technical support, resolving complex ticket escalations and ensuring operational continuity.
- Managed network infrastructure, troubleshooting outages and optimizing performance for domain connectivity.
- Configured hardware assets (laptops, mobile devices), ensuring strict adherence to security protocols.

Kenya National Library Service (KNLS) | Volunteer & Trainer | Jan 2024 – Mar 2024

- Led digital literacy workshops, teaching computer proficiency to students and community members.
- Assisted in organizing educational workshops and fostering a collaborative learning environment.
- Assisted in front desk services and library resource organization.

SELECTED ENGINEERING PROJECTS

AI & Machine Learning

- **Meeting Intelligence Engine (AI + Real-time Systems):** An end-to-end platform that captures meetings, produces live transcripts with speaker labels, and generates AI summaries, action items, and searchable chat. Includes calendar integration, secure access controls, and auto-generated weekly audio recaps. Built with GPT-4o on a React and Python stack.
- **DeepDetect (AI/ML):** An explainable deep-learning system for detecting deepfake images and video, trained on a large public dataset with strong accuracy. Provides visual explanations of its predictions, video analysis, and admin reporting. Built with a ResNet50 model, React/TypeScript, and Flask.
- **Gaming RAG Assistant (AI):** A question-answering assistant over game documentation that combines semantic and keyword search for accurate, character-styled responses. Built with HuggingFace models and deployed on Streamlit Cloud.

Data & Platforms

- **Corridor Intelligence Platform (Verst Carbon):** A decision-support platform for the Abidjan-Lagos Corridor, a 1,028 km highway corridor linking five West African nations. Consolidates infrastructure, economic, policy and stakeholder data into one dashboard, with an interactive map and a chat-driven agent that answers questions and drives contextual map overlays. Built with Next.js, React and TypeScript for the EU-Africa Business Forum.
- **cleancookIQ (Fullstack, co-built with Ignis Innovation):** A platform coordinating Kenya's institutional shift away from polluting cooking fuels, giving institutions, suppliers, funders and researchers each their own workspace. Covers carbon project registration, credit verification and digital monitoring records alongside a supplier marketplace and financing tools. Built with React, TypeScript and Supabase.
- **Soil Health Dashboard (Geospatial Data):** An analytical dashboard unifying authoritative soil datasets into one workspace. Queries live soil properties for any coordinate, compares sources against each other, and scores land degradation risk across a point or a drawn polygon, carrying uncertainty ranges through to the interface. Built with Next.js, TypeScript and MapLibre GL.
- **Smart Farming Assistant (IoT + AI):** A farm management dashboard combining live sensor readings for soil moisture and environmental conditions with weather forecasting and an AI assistant that answers agronomic questions. Built with React, TypeScript and FastAPI.

Mobile & Web

- **CasaControl (Mobile):** A smart-home management mobile app built with Kotlin, enabling remote appliance control.
- **CommunityPlus (Web):** Web-based donation platform bridging donors and receivers to facilitate community support.
- **Client Portfolio Websites (Freelance):** Designed and delivered personal portfolio sites for three professional clients, covering responsive layout, content structure and deployment.

ACHIEVEMENTS & CERTIFICATIONS

- **Dean's List Award (2024-2025, 2025-2026):** Awarded for maintaining a high GPA and demonstrating academic leadership.
- **NSK AI RAG Bootcamp:** Completed intensive training on building Retrieval-Augmented Generation systems.
- **Cisco Certified:** IT Essentials (Cisco Networking Academy) – proficient in hardware assembly, maintenance, and security.
- **Cisco Certified:** Computer Networks (Cisco Networking Academy).

LEADERSHIP & ACTIVITIES

- **Local Committee Vice President** – Finance & Legalities, Strathmore University.
- **Project Manager** – SDG Hub, Strathmore University.
- **Member:** Google Student Developer Club (GDSC), Strathmore University.
- **Member:** Strathmore Computing and Engineering Sciences Association.
- **Registered Member:** The Kenya Red Cross Society.